PROJECT REPORT

Journal-entry testing, **decided.**

JE Agent — Journal Entry Testing AI Agent
per ISA 240 / AS 2401

PREPARED BY

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Journal Entry Testing, Decided.

JE Agent, an AI-assisted audit assistant for journal entries (ISA 240 / AS 2401)

This report presents JE Agent, a software tool that assists auditors in journal-entry testing under ISA 240 and AS 2401. It describes the problem the tool addresses, the rationale behind its architecture, and the results of a controlled evaluation on populations of up to 200,000 lines. The document is written for a non-technical reader; implementation detail is available in the repository documentation (`DESIGN.md`, `PROJECT_REPORT.md`).

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1 · The Problem

ISA 240 and AS 2401 require the auditor to examine journal entries for indications of management override of controls. In practice this examination faces a scale problem. A full-year export commonly contains tens or hundreds of thousands of entries, and the population of genuinely relevant entries is small.

Current practice handles this in three ways, each with a known weakness.

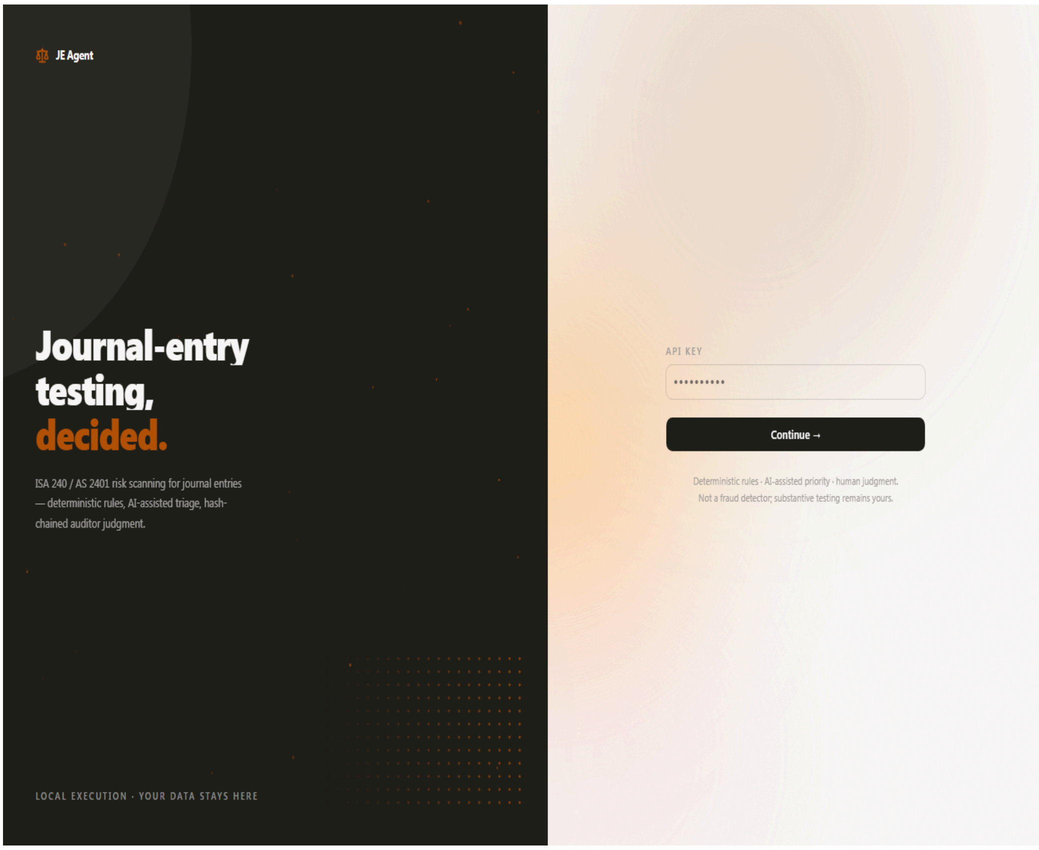
Manual review does not scale. Reading a complete ledger is not feasible, so work is performed on samples; entries outside the sample are never examined, and the coverage actually achieved is hard to state.

Scripted filters scale but produce noise. A criterion such as "round amounts above a threshold" returns large result sets dominated by legitimate postings: commercial invoices are frequently round, and posting near period end is ordinary practice. The reviewer's effort is spent dismissing false positives, and the effective quality of the screen is unknown.

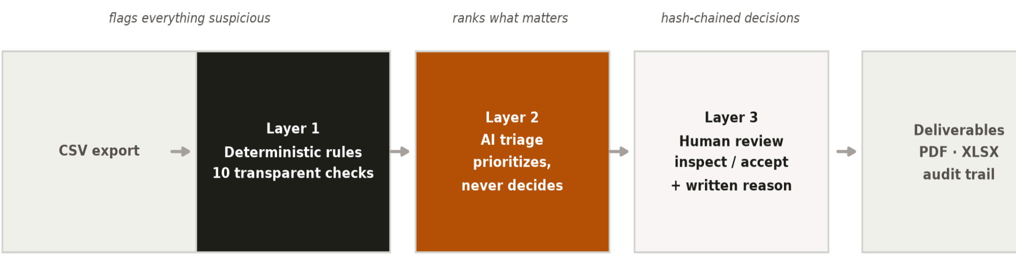
AI-only analysis introduces an accountability gap. A language model can classify entries, but it cannot substantiate its output to the standard required for audit evidence, and its conclusions cannot be defended in a file review without traceable reasoning.

These weaknesses are complementary rather than cumulative. Manual review provides judgment at low volume, scripts provide coverage without judgment, and models provide prioritization without accountability. JE Agent is an attempt to combine the three so that each operates only where it is reliable.

2 · The Logic, Three Layers



JE Agent processes an accounting export through three successive layers.



Layer 1, deterministic rules. Ten transparent checks examine every entry: round amounts, splitting, period-end timing, balance failures, rare users, unusual account pairings, reversals, date divergence, high-risk users, and manual-entry concentration.

Each check is a readable, inspectable condition that states why it fired. Rules provide complete coverage of the patterns they encode, but they also flag many legitimate postings that share surface characteristics with risk indicators. Their role is screening, not conclusion.

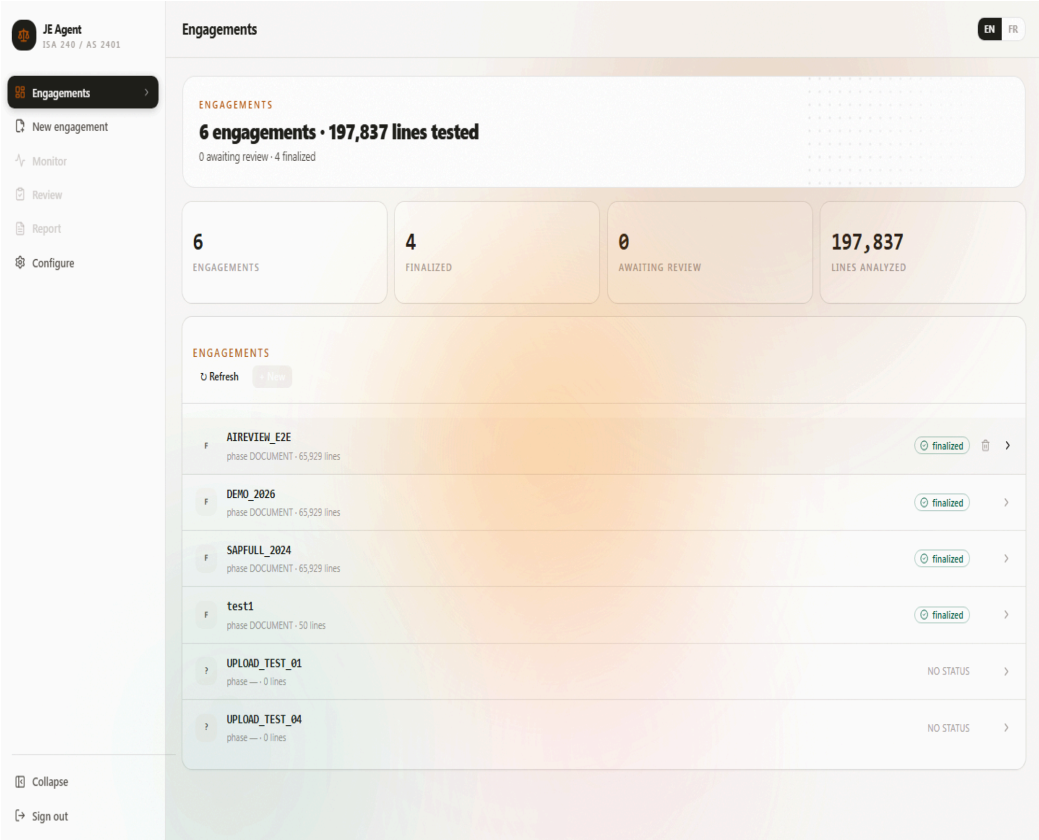
Layer 2, AI triage. A language model ranks the flagged entries by likely relevance and produces a short written rationale for each ranking. Its output is recorded as machine-generated advice and carries no decision authority. This layer reduces the order in which a human reviews material; it does not reduce the human's responsibility for what is decided.

Layer 3, human judgment. Every entry that reaches review requires an explicit auditor decision, inspect or accept, supported by a mandatory written reason. Decisions are recorded in a hash chain: each record contains the fingerprint of its predecessor, so any later modification of the trail is detectable.

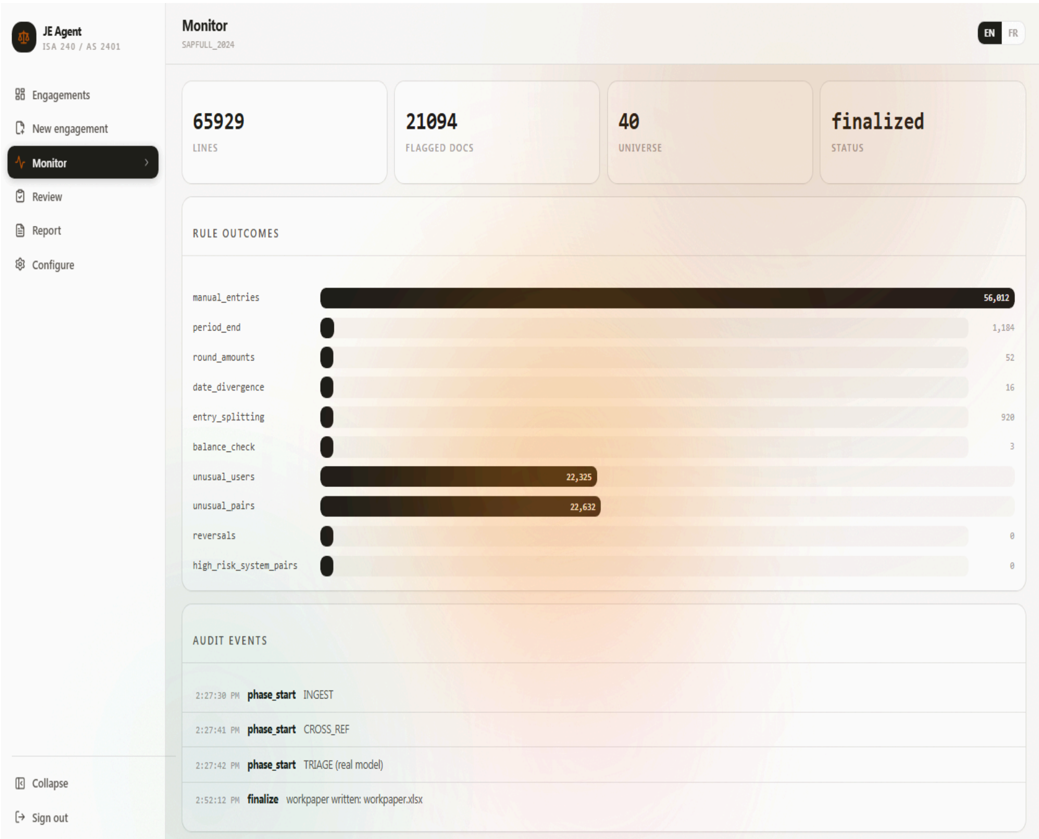
On completion the pipeline produces the engagement deliverables: a report in PDF form (English or French), an Excel workpaper, and the flagged-entry listing.

3 · The Console

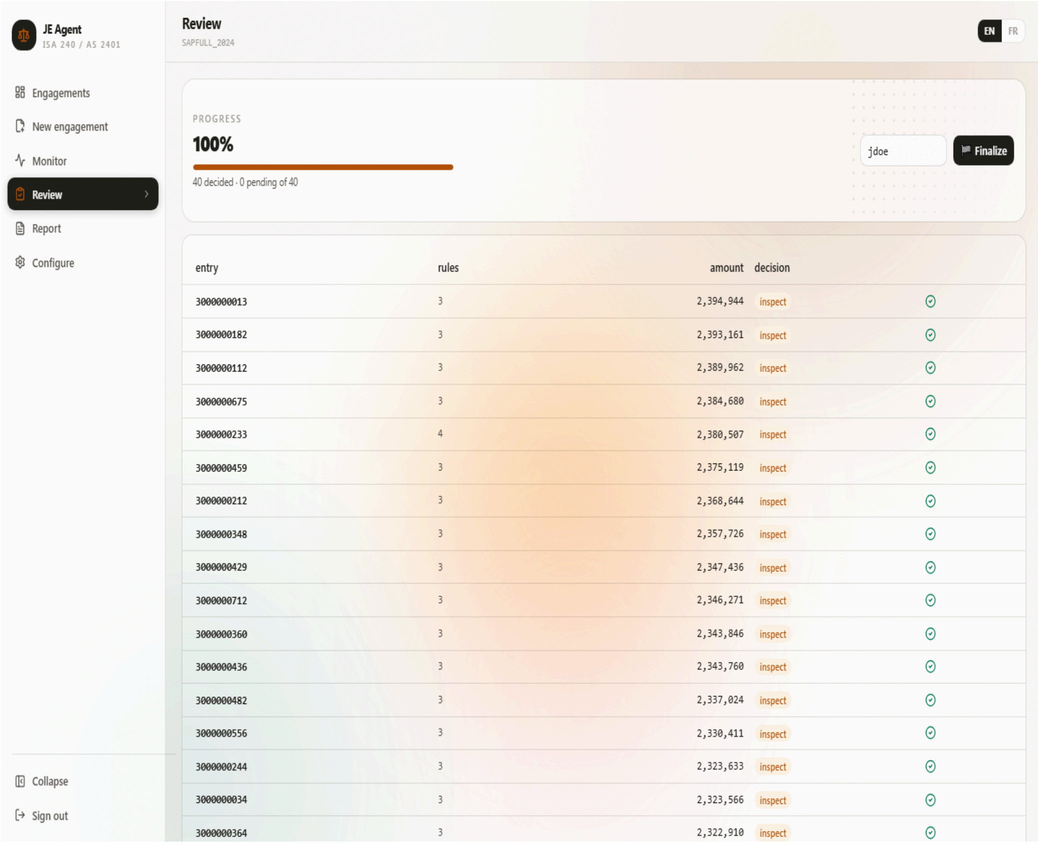
JE Agent runs locally. Engagement data remains on the host machine; only optional triage calls leave it, and those operate in zero-retention mode.



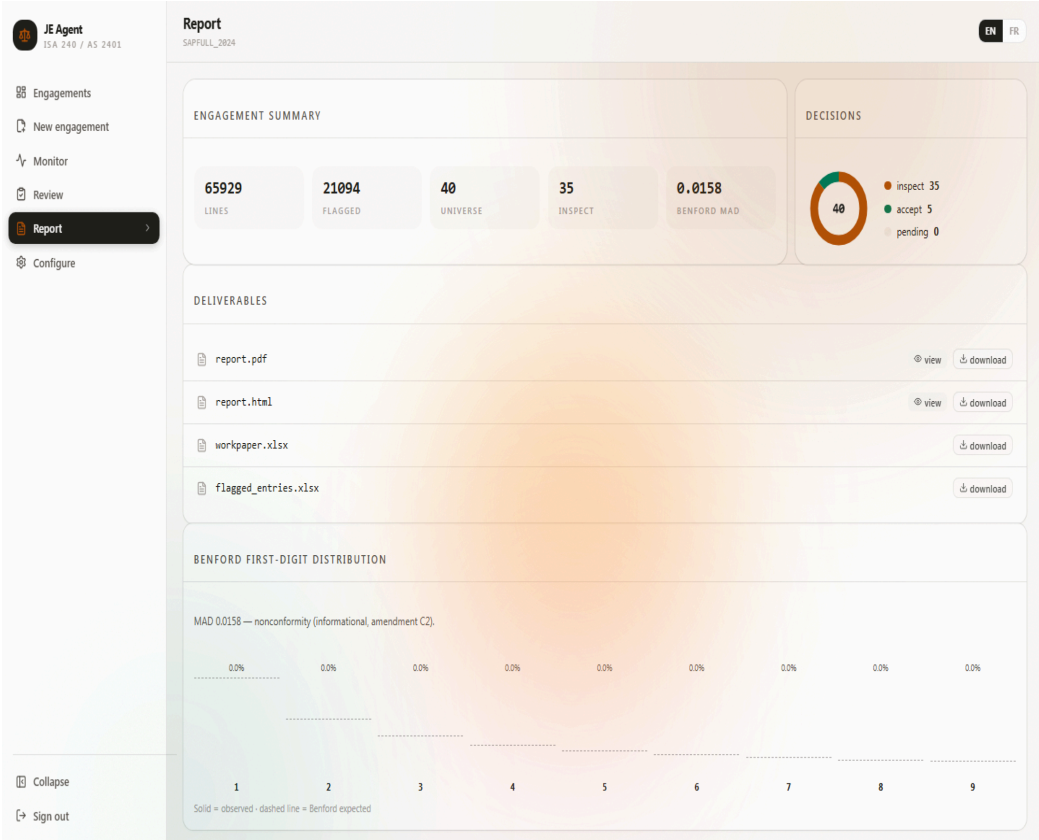
The dashboard lists every engagement with its population, phase, and status.



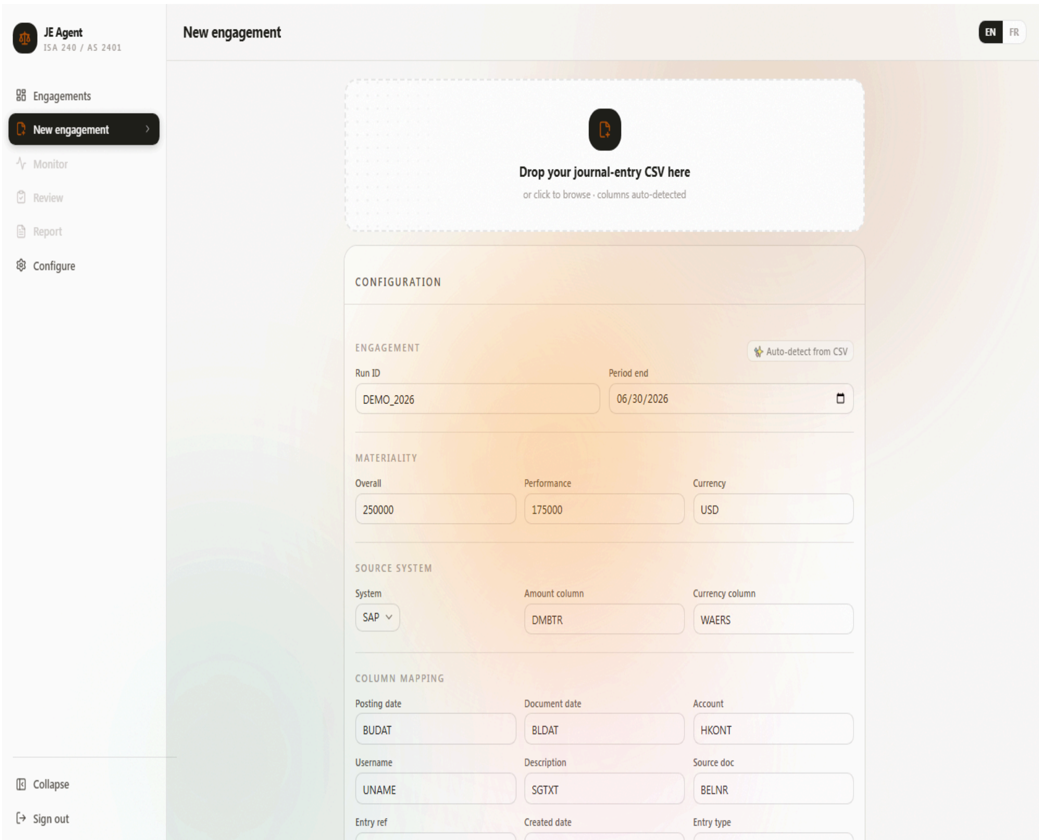
Monitor shows per-rule flag counts and the pipeline event timeline for the selected run.



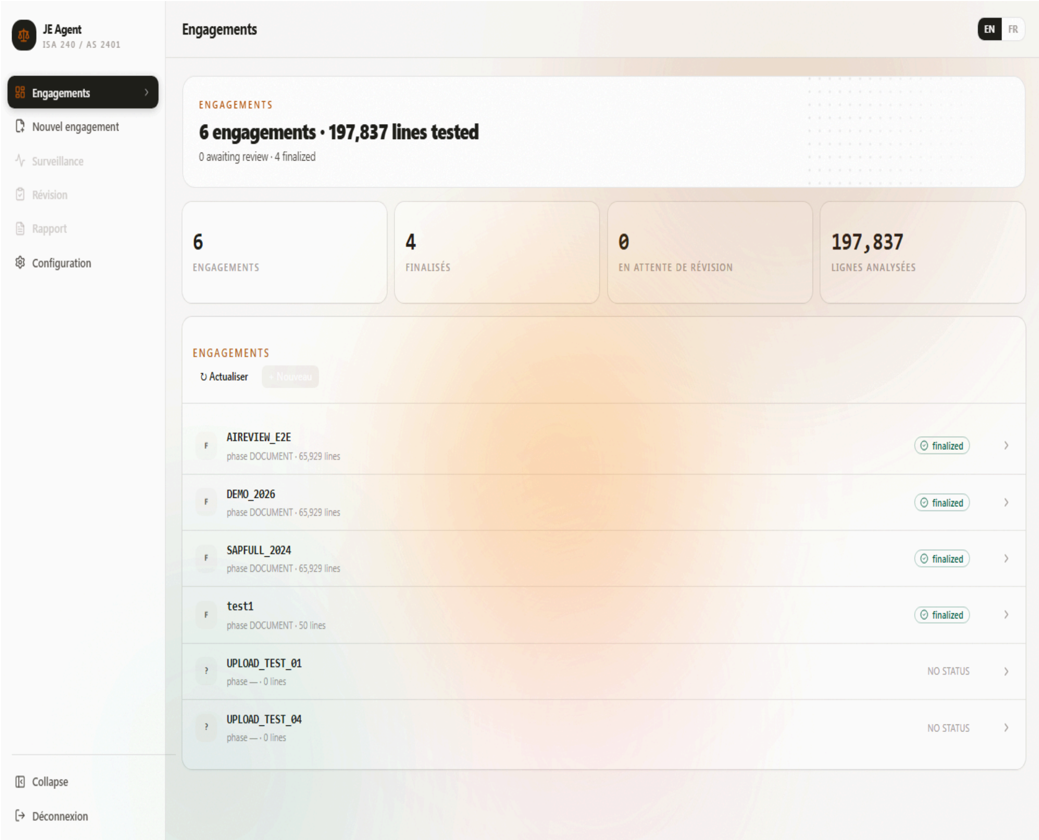
The review queue presents ranked entries with the rules that fired. Selecting an entry opens its journal lines. Recording a decision requires one action plus a written reason.



The Report page collects deliverables, summarizes decisions, and reports first-digit frequencies against the Benford reference as an informational indicator.



An engagement starts with a CSV upload. Column mapping is detected automatically; materiality, review mode, and report language are set in one form.



The interface and the generated report are available in English and French.

4 · Evaluation

The purpose of the evaluation was to measure detection performance against known ground truth. Real ledgers do not come labeled, so the evaluation used synthetic populations into which anomalies were injected deliberately. Every injected anomaly carries a label naming the rule expected to detect it, which allows recall and precision to be computed exactly.

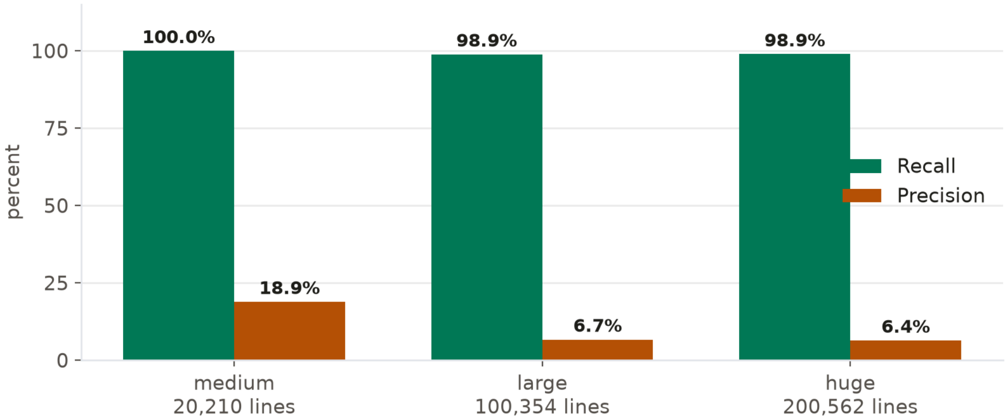
Methodology

Three populations were generated, ranging from 20,210 to 200,562 lines, each deterministic under its seed. The clean base was constructed to resemble operational data rather than an idealized one: approximately 90% manual and 10% system-posted entries, users with legitimately low activity, legitimately large invoices, and month-start postings. This noise matters, because it allows false positives to occur and therefore to be counted.

For each rule, recall is the share of injected anomalies detected, and precision is the share of flags corresponding to an injected anomaly. Flags raised on clean documents count as false positives.

Results by scale

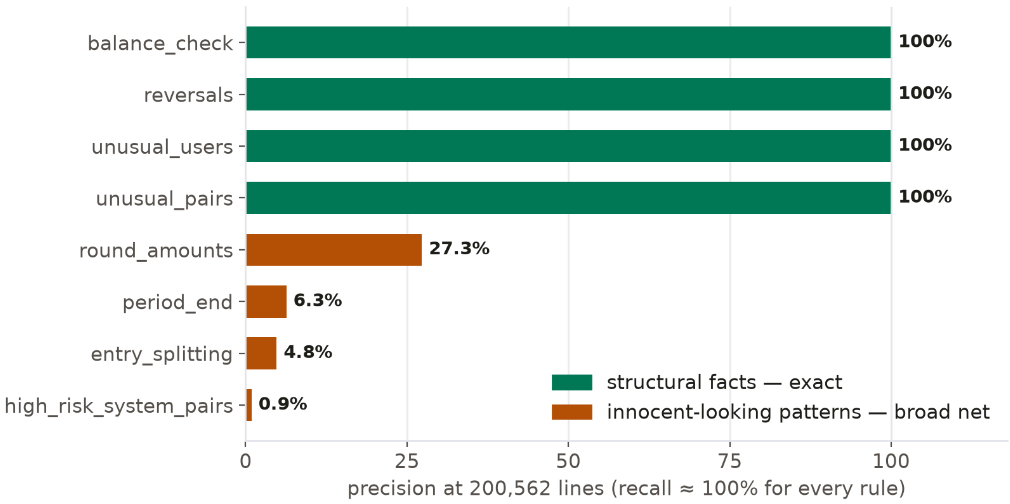
Scenario	Lines	Injected	Recall	Precision
medium	20,210	105	**100%**	18.9%
large	100,354	177	**98.9%**	6.7%
huge	200,562	281	**98.9%**	6.4%



Recall remains at or near 100% at every scale: almost no injected anomaly escapes detection, and the largest population ingests in about fourteen seconds. Precision declines as population grows, because the frequency of benign entries that resemble risk indicators grows with the population, while genuine anomalies remain rare by nature. Both observations are expected consequences of the design intent: layer one screens completely and accepts noise in exchange.

Per-rule results (200k-line scenario)

Rule	Recall	Precision
round_amounts	100%	27%
entry_splitting	100%	5%
period_end	100%	6%
unusual_pairs	96%	100%
unusual_users	100%	100%
balance_check	100%	100%
reversals	100%	100%
date_divergence	87%	100%
high_risk_system_pairs	100%	0.9%



Results separate into two groups. Rules defined over structural facts, failed balance checks, true reversal pairs, users with negligible history, account pairs absent from the baseline, achieve near-perfect precision because such facts have no common innocent explanation. Rules defined over statistical tendencies, roundness, timing, size clustering, retain full recall but with declining precision, since legitimate entries share these characteristics. The distinction informs how each rule's output should be weighted in review.

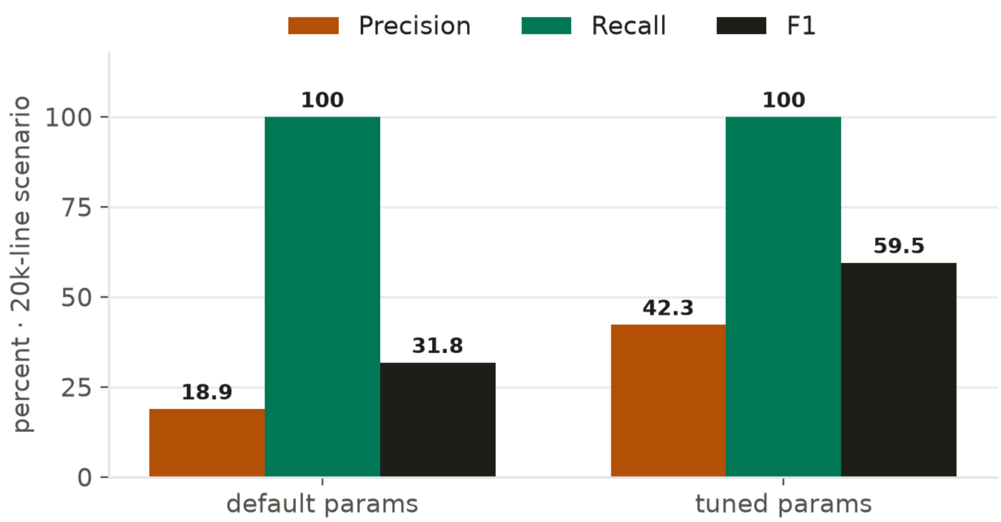
Universe scoping

Reviewers see a capped, ranked selection (the universe) restricted to entries above performance materiality. Between 97% and 100% of above-materiality injected anomalies reached the universe in testing. Sub-materiality exclusion behaves as specified.

Parameter sensitivity

Rule thresholds were treated as configurable hypotheses rather than constants. On the medium scenario, two parameters were adjusted (a narrower period-end window and a stricter rarity threshold for system accounts) and the run repeated.

Metric	Default	Tuned
Aggregate precision	18.9%	**42.3%**
Aggregate recall	100%	**100%**



Precision improved by a factor of 2.2 with no loss in recall. Because thresholds interact with population composition, they should be calibrated against each engagement's own materiality; the harness makes that calibration measurable.

Triage layer

The triage model was evaluated separately on the smallest scenario. It identified 100% of injected anomalies, with inspect precision of about 41%. The model therefore performs its intended function: broad prioritization ahead of human review. Precision is achieved by the reviewer, not the model, which is consistent with the allocation of responsibility described in Section 2.

5 · Trust and Integrity

Four mechanisms protect the integrity of the working papers.

Hash-chained decisions. Each decision record includes the fingerprint of the previous record. Modification of any historical entry breaks the chain and is detectable.

Citation gates. The generated narrative may not be finalized unless each factual claim cites data present in the run.

Deterministic core. Identical input produces identical flags, allowing results to be reproduced at any later date.

Human-only finalization. Four completion gates (review coverage, documented procedures, valid citations, stated limitations) must pass before finalization. Finalization cannot occur without recorded human decisions.

6 · Limitations

The following limitations were observed directly in the evaluation.

Aggregate precision at scale is low by construction: 6.4% at 200k lines. The rule layer intentionally favors completeness over selectivity, and the reviewer's workload reflects that choice. Users should expect to review flags, not receive verdicts.

`high_risk_system_pairs` depends on configuration. High-risk users must be declared for the rule to operate meaningfully; in the unconfigured state it measured 0.9%

precision.

`date_divergence` missed roughly 13% of injections whose falsified document dates fell inside the accepted window.

The evaluation populations, although textured to resemble operational data, remain synthetic. Behavior on live client data will differ in ways the harness cannot fully anticipate.

7 · Conclusion

JE Agent combines deterministic screening, AI-assisted prioritization, and mandatory human judgment for journal-entry testing under ISA 240 / AS 2401. The evaluation supports three conclusions. Detection of injected anomaly classes is near-complete and scales to 200,000 lines within seconds. The architecture assigns final authority exclusively to the reviewing auditor. And the system's limitations were measured and published alongside its strengths, which is the standard its own audit trail is designed to enforce.

Appendix A · The ten rules in plain language

#	Rule	What it looks for	Why it matters
1	manual_entries	entries typed by hand rather than posted by a system	manual entries are where override risk lives
2	round_amounts	amounts on suspiciously round numbers (10,000 / 100,000...)	invented figures are rarely precise
3	entry_splitting	one obligation broken into pieces just under approval	classic way to dodge a second signature

#	Rule	What it looks for	Why it matters
		limits	
4	period_end	entries posted in the window around period close	closing pressure is when numbers get “managed”
5	unusual_pairs	account combinations that rarely occur together	e.g. revenue debited against an asset account
6	unusual_users	user IDs with almost no posting history acting alone	ghost or borrowed credentials leave little trace
7	balance_check	documents whose debits and credits don’t reconcile	unbalanced postings indicate broken controls
8	reversals	an entry reversed shortly after posting, same amount	reversal after benefit = possible concealment
9	date_divergence	document date far from posting date	backdating shifts transactions into wrong periods
10	high_risk_system_pairs	configured high-risk users combined with unusual accounts	privileged users need scrutiny wherever they touch

Every rule is a transparent, inspectable check. No black boxes operate at this layer.

Appendix B · Running it yourself

JE Agent runs entirely on your machine. Three commands from a fresh clone:

```
git clone https://github.com/salaheddine-11/je-agent && cd je-agent
uv sync
uv run jeagent serve
```

Then open `http://localhost:8300` in a browser (default key: `jeagent`).

A typical engagement takes about five minutes of attention:

1. **New engagement**: drop in your CSV export; column mapping is detected automatically. Set overall/performance materiality, choose human or AI-assisted review mode, pick report language.
2. **Monitor**: watch the pipeline stages verify themselves: ingest → rules → cross-reference → triage.
3. **Review**: work the ledger. Every entry shows which rules fired and its full journal lines; each decision needs one click plus a written reason.
4. **Finalize**: four gates check review completeness, procedures, citations and limitations; then the PDF report, Excel workpaper and flagged-entry list are generated.

Requirements: any modern OS, Python 3.12+, ~200 MB free space. Internet is needed only for the optional AI triage calls (zero-retention mode); everything else runs locally, so client data never leaves the machine.

Glossary

Journal entry (JE): the atomic record of an accounting transaction: dated lines, each crediting or debiting an account.

ISA 240 / AS 2401: international (IAASB) and US (PCAOB) standards requiring auditors to address fraud risk, explicitly including journal-entry testing.

Materiality: the size threshold above which misstatements matter to the financial statements. JE Agent distinguishes *overall* materiality from *performance* materiality (the working threshold for testing).

Performance materiality (PM): see above; entries above PM drive the review universe's scope.

Universe: the ranked selection of flagged entries the reviewer actually sees, scoped to items above performance materiality and capped for practical review.

Recall: of all true anomalies present, the share the system caught. Missing fraud is the costly error, so recall is the headline metric.

Precision: of everything the system flagged, the share that was genuinely anomalous. Low precision means noise the reviewer must wade through.

F1: the balanced average of recall and precision.

Benford's law: natural data has predictable first-digit frequencies (1 is far more common than 9). Fabricated numbers drift away from this pattern; JE Agent reports the comparison as an informational indicator only.

Hash chain: each decision record carries a cryptographic fingerprint of the previous one, so altering history breaks the chain and becomes detectable.

Zero retention: AI calls configured so the provider stores nothing about the request contents.